Here we agree that X is linked to itself (by a chain of length 0).

This linking relation is clearly an equivalence relation, so it defines a splitting of the set S of isomorphism classes of simple A-modules into equivalence classes S_k, $k \in B$. The kth **block** $\mathcal{C}_k$ of $\mathcal{C}$ is, by definition, the category of all objects M of $\mathcal{C}$ such that all simple modules occurring in the Jordan-Hölder series of M are in S_k.

Example 9.5.2. If A is semisimple, each block is equivalent to the category of vector spaces (and thus has only one simple object). If A is a commutative local artinian algebra, then there is only one block. Also there is just one block for the algebra of Problem 9.3.2.

Problem 9.5.3. (i) Show that there is a natural bijection between blocks of $\mathcal{C}$ and indecomposable central idempotents e_k of A (i.e., ones that cannot be nontrivially split in a sum of two central idempotents), such that $\mathcal{C}_k$ is the category of finite dimensional e_kA-modules.

(ii) Show that any indecomposable object of $\mathcal{C}$ lies in some $\mathcal{C}_k$ and that $\mathrm{Hom}(M, N) = 0$ if $M \in \mathcal{C}_k$, $N \in \mathcal{C}_l$, $k \neq l$. Thus, $\mathcal{C} = \bigoplus_{k \in B} \mathcal{C}_k$.

(iii) Determine the blocks in the category of left A-modules for $A = k[S_3]$, where k is of characteristic 2.

9.6. Finite abelian categories

Let $\mathcal{C}$ be a k-linear abelian category in which any object has finite length (i.e., a finite filtration whose successive quotients are simple). Assume that for every simple object X of $\mathcal{C}$, one has $\mathrm{End}(X) = k$. Note that in such a category, Hom spaces are finite dimensional (check it!).

Definition 9.6.1. Let us say that $\mathcal{C}$ is **finite** if it has enough projectives and finitely many simple objects (up to an isomorphism).

For example, as we have shown, the category of finite dimensional modules over a finite dimensional algebra is finite. Below we will see that the converse is also true.

Definition 9.6.2. An object P of an abelian category $\mathcal{C}$ is said to be a **projective generator** (or **progenerator**) if it is projective and every object is a quotient of a multiple of P.